

HITESH TOMAR

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EDUCATION

University School of Automation and Robotics (USAR), GGSIPU

Delhi, India

Bachelor of Technology in Artificial Intelligence and Machine Learning (CGPA: 8.8/10)

2024 – 2028

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Operating Systems, DBMS, Computer Networks

TECHNICAL SKILLS

Languages: Python (3.11+), C++, JavaScript (ES6+), SQL, HTML/CSS

Machine Learning & Edge AI: TinyML, TensorFlow Lite Micro, Scikit-learn, XGBoost, INT8 Quantization, SciPy, SHAP

AI & LLM Systems: LangGraph (Multi-Agent), RAG Pipelines, Google Gemini API, Prompt Engineering, Agentic AI

Backend, DB & Cloud: FastAPI, Flask, Node.js, Express, PostgreSQL, SQLite, Prisma ORM, Docker, Power BI, GEE, QGIS

EXPERIENCE

AICTE IDEA Lab, GGSIPU

New Delhi, India

Machine Learning Engineer Intern – Edge AI & TinyML

Summer 2026

- Engineered TinyML vital-monitoring pipeline for “NDRF Guardian”, deploying an INT8 model quantized to **5.8 KB** via TFLite Micro for **~7ms inference** on ESP32-S3 with LoRa SX1278 (5–10 km) in dead-zones.
- Trained dual-output multi-task ANN predicting risk levels and continuous safety index with **95% accuracy** and **96% critical recall** on a synthesized 15,000-sample clinical dataset modeled on AHA, WHO, and NIOSH thresholds.

Bluestock™

Delhi, India

Data Analyst Intern – FinTech

June 2026 – Aug 2026

- Architected financial ETL pipeline ingesting live NAV feeds across 40+ mutual fund schemes into an optimized SQLite data warehouse.
- Implemented quantitative financial risk models computing CAGR, Sharpe, Sortino, Alpha, Beta, VaR, and CVaR, visualized via an interactive 5-page Power BI executive dashboard.

India Space Academy (ISA)

Delhi, India

Remote Sensing & GIS Intern

Jan 2025 – Feb 2025

- Developed an automated surface water extraction pipeline using Sentinel-2 multispectral imagery in Google Earth Engine (GEE) to monitor the Yamuna River basin.
- Calculated NDWI across Green and NIR bands, applying focal-mode filtering and QGIS vector analysis to quantify **6.90 km²** of water surface area.

PROJECTS

EcoScout – Multi-Agent Sustainability Analyst | *LangGraph, Python, FastAPI, Gemini, SSE* | [\[GitHub\]](#)

- Architected a multi-agent LangGraph system for UN SDG 12 executing parallel web research and automated product sustainability scoring (0–100).
- Coordinated concurrent domain agents (Materials, Supply Chain, Packaging) using live Tavily search and Gemini synthesis, streaming thought steps to React over SSE.

SourceMate AI – Multi-Source Research Agent | *React, Node.js, Express, Gemini API, SSE, Docker* | [\[GitHub\]](#)

- Built document workspace and research chatbot streaming context-grounded Gemini responses over Server-Sent Events (SSE).
- Engineered browser IndexedDB caching for large PDF source uploads, eliminating server file overhead; containerized full-stack services with Docker.

Digital Footprint Intelligence Platform | *Python, Flask, NetworkX, GeoPy, React, Socket.IO* | [\[GitHub\]](#)

- Built digital forensics platform correlating Call Detail Records (CDR), cell tower locations, and banking transactions to detect fraud rings.
- Implemented 7 graph and statistical anomaly algorithms uncovering money laundering layering ($A \rightarrow B \rightarrow C$), rendered via dynamic NetworkX topologies and real-time Socket.IO.

ACHIEVEMENTS & CERTIFICATIONS

Smart India Hackathon (SIH) 2025: Qualified through multiple evaluation rounds with AI-powered healthcare triage solution.

Problem Solving: Solved **200+ Data Structures & Algorithms** problems across LeetCode and GeeksforGeeks.

Certifications: Data Structures and Algorithms with C/C++ (Wipro Pregrad) · Remote Sensing and GIS (India Space Academy)